



AstraPulse

Mission Readiness & Predictive Maintenance Copilot

From sensor data to mission decision — in one pipeline.
Team KMstra | Track: AI | IBM Bob AI Hackathon



Disconnected Signals



Sensor & HUMS data contains early degradation signals but remains siloed — disconnected from failure risk, RUL estimates, and mission requirements.

Reactive Decisions



Manual, reactive maintenance creates compounding risks: safety incidents, unplanned downtime, and escalating operational costs across mission-critical fleets.



No Readiness Score

Without a unified readiness signal, operators cannot assess true mission readiness — every decision is a guess under pressure with incomplete information.



The Problem

The Solution



IsolationForest

Detects anomalies and scores asset health from raw sensor streams. Flags early degradation signals before failure thresholds are reached.



XGBoost

Predicts failure risk probability using engineered features. Classifies assets as Low, Medium, or High risk based on sensor patterns.



RandomForest

Estimates Remaining Useful Life (RUL) in cycles. Provides time-to-failure horizon to inform scheduling and mission planning.

0-100

Mission Readiness Score

4-Tier

Maintenance Priority

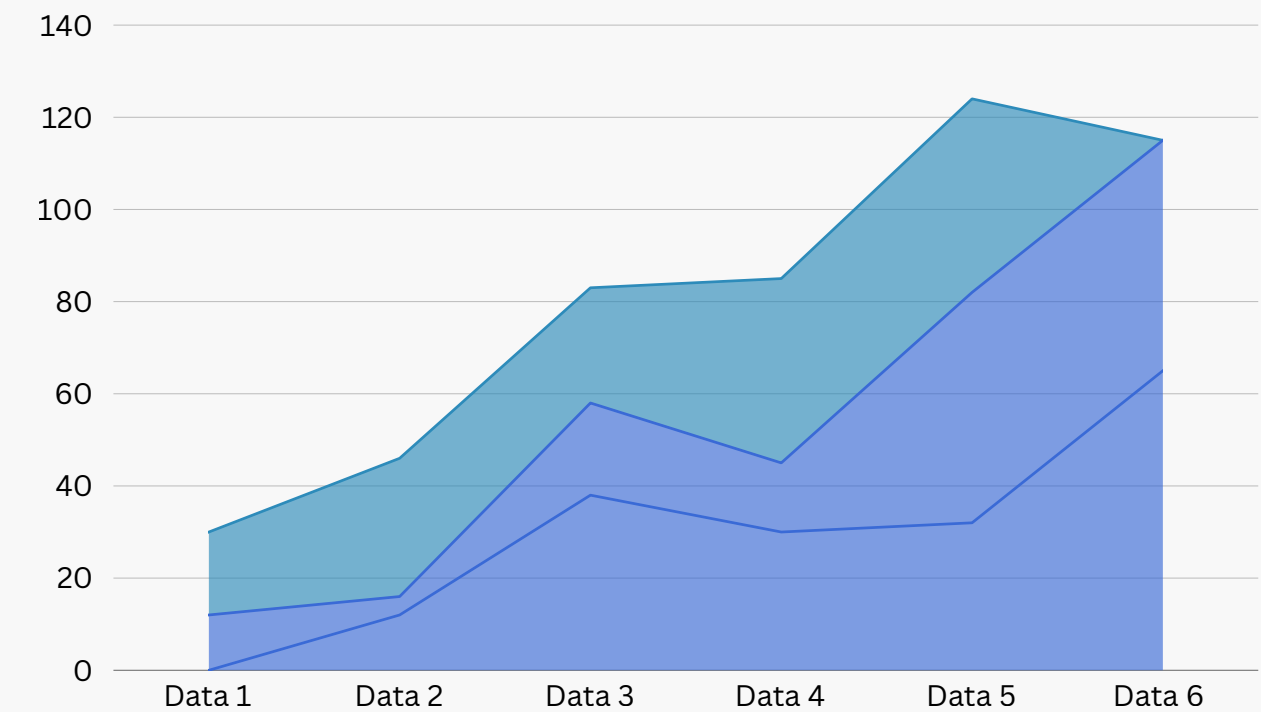
3-Way

Asset Status: READY /
ADVISORY / NOT READY

System Architecture

End-to-End Pipeline

- NASA C-MAPSS FD001 raw sensor data → Preprocessing (17 features + RUL labels) → Feature Engineering (85 features) → Train 3 ML Models (IsolationForest, XGBoost, RandomForest) → Readiness Engine → asset_readiness.csv
- Runtime: FastAPI backend → React Mission Intelligence UI. IBM Bob connects via MCP with 5 operator tools for fleet readiness, asset status, time series, maintenance priorities, and mission readiness decisions.



Live Demo Walkthrough



Step 1: Upload

Upload multi-cycle CSV from an unseen engine. System validates, cleans data, and engineers 85 features automatically.



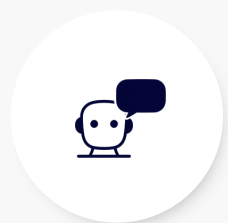
Step 2: Assess

POST /api/assess runs 3 ML models: IsolationForest health, XGBoost failure risk, RandomForest RUL on latest engine cycle.



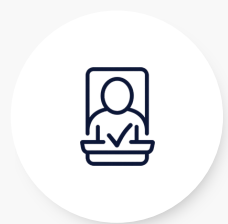
Step 3: Output

Frontend displays Health Score, Failure Risk, RUL, Readiness Status, and maintenance priority in real time.



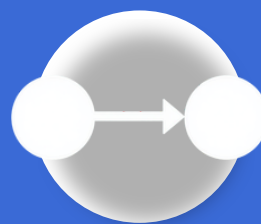
MCP Tools

5 operator tools via MCP: fleet readiness, asset status, asset time series, maintenance priorities, and mission readiness. Structured context handoff.



Operator Layer

Bob converts ML model outputs into plain-language operator decisions. It is not the predictive model — it interprets and communicates results.

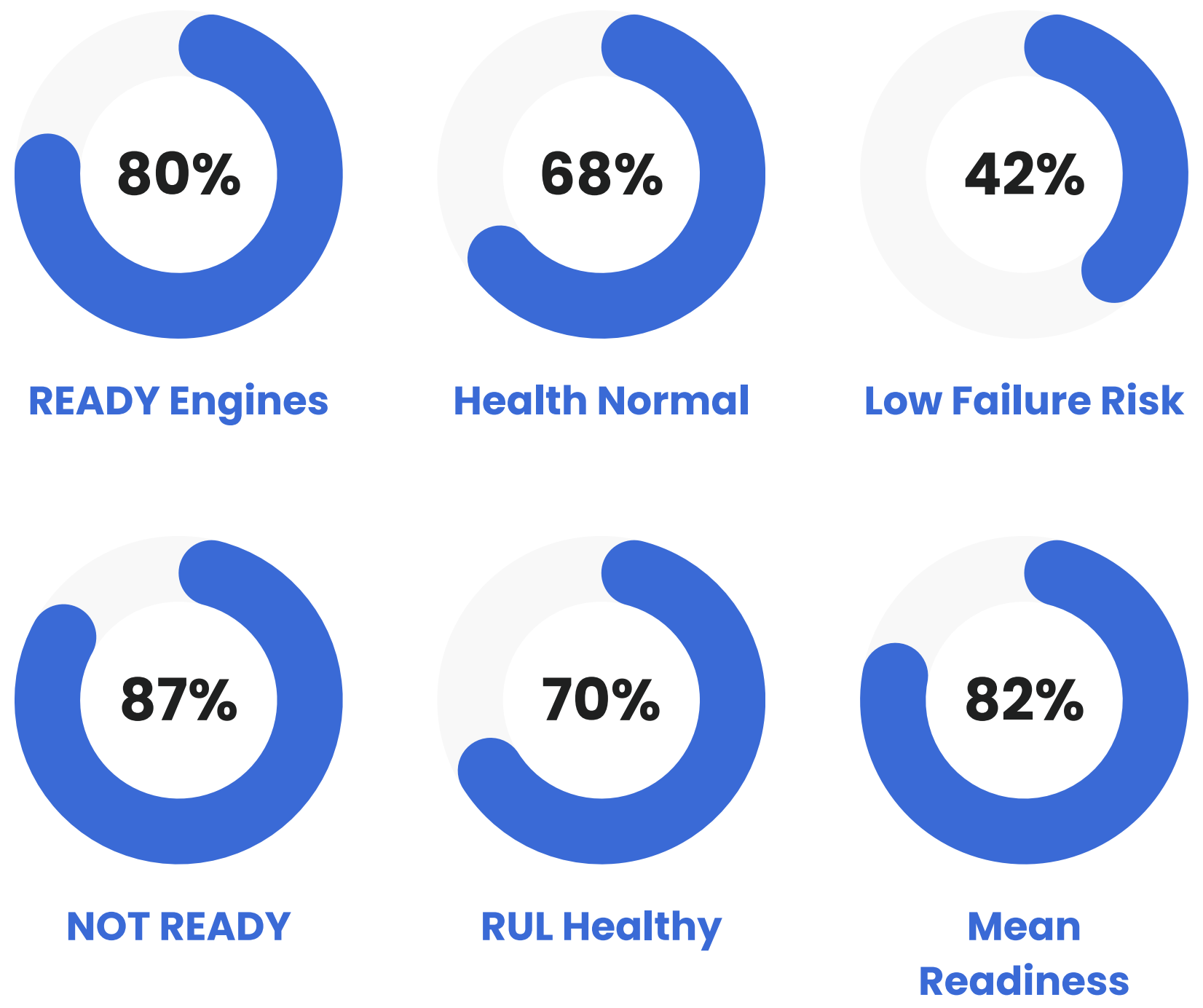


Context Handoff

Structured context from readiness engine is passed to Bob. Each query returns actionable guidance: go/no-go, maintenance urgency, and fleet status.



IBM Bob Copilot



Results & Impact

- 100 test engines assessed: 70 READY, 10 ADVISORY, 20 NOT READY.
- Mean readiness score: 73.2/100.
- High-risk alerts: ENG-034 and ENG-024 flagged for immediate action.
- Maintenance priorities: 12 immediate, 10 high, 8 medium, 70 low.
- Health distribution: 57 normal, 22 warning, 21 critical.
- Failure risk: 78 low, 4 medium, 18 high. RUL classification: 44 healthy, 56 advisory.
- AstraPulse delivered end-to-end readiness assessment across all 100 unseen engines with zero manual intervention.

Team KMstra

- Vaibhavi Kariya (ML + Integration)
- Pushti Kansara (ML Health)
- Disu Makadiya (ML Failure)
- Rutvi Shah
- Shared Build: NASA C-MAPSS Pipeline · Synthetic Data · 3 ML Models · Readiness Engine · FastAPI + React · IBM Bob MCP · Automated Testing + Frontend + Data

Sensor Data → Health → Failure Risk → RUL → Readiness → Mission Decision